

# CROP HEALTH MONITORING USING AI

## *Smart Farming for a Sustainable Future*

### **1. Introduction**

Artificial Intelligence (AI) is transforming agriculture into a smart and efficient system. Crop health monitoring using AI enables early detection of plant diseases before visible symptoms appear. This approach helps farmers take preventive measures, improve productivity, and reduce resource wastage.

### **2. Problem Statement**

Traditional farming faces several challenges such as late disease detection, excessive use of fertilizers and pesticides, manual monitoring, and unpredictable environmental conditions. These issues lead to lower yield and increased costs.

### **3. Objectives**

The main objectives include developing an AI-based crop monitoring system, optimizing resource usage, enhancing productivity, and enabling farmers to make data-driven decisions.

## **4. Methodology**

The study uses both quantitative and qualitative methods. Quantitative data includes satellite imagery, IoT sensor data, and performance metrics. Qualitative methods include farmer feedback, case studies, and comparative analysis between traditional and AI-based farming.

## **5. Data Collection**

Data is collected from multiple sources such as satellite and drone imagery, soil sensors measuring moisture and nutrients, weather data including temperature and rainfall, and crop growth tracking systems.

## **6. AI Framework**

The AI system works in four stages: Observe, Analyze, Predict, and Act. Technologies such as computer vision, machine learning, and predictive analytics are used to detect diseases, predict crop yield, and suggest actions.

## **7. Results**

AI-based crop monitoring has shown 95% detection accuracy, 30% reduction in pesticide usage, and 20% increase in crop yield. It enables early disease detection and precise irrigation planning.

## **8. Conclusion**

AI transforms agriculture from a reactive to a proactive system. It improves efficiency, reduces environmental impact, and supports sustainable farming practices.

## **9. Recommendations**

Adoption of affordable AI tools, improved rural infrastructure, farmer training programs, and government support are essential for large-scale implementation.

## **10. Future Scope**

Future advancements include integration with IoT devices, drones, and advanced analytics for even more accurate and efficient farming systems.